

Paulo Ricardo Marques de Araujo

☎ (343) 580-2041

✉ paulo_marques_araujo@hotmail.com

🌐 paaraujo

🔗 PRM de Araujo

SUMMARY

Senior Researcher and Technical Lead with 15+ years of experience in Autonomous Systems and Digital Twins. Expert in designing and deploying complex sensor fusion pipelines and AI-driven state estimation. Proven track record in leading cross-functional teams to validate algorithms on real-world vehicle platforms. Creative problem-solver with a entrepreneurial mindset, dedicated to advancing robust navigation in challenging, real-world environments.

CORE COMPETENCES

- **Sensor Fusion & Estimation:** Invariant Kalman Filters (IEKF), EKF/UKF, Multi-sensor Fusion (Radar, LiDAR, GNSS, IMU, Odometer, Camera), SLAM, and Vehicle Localization.
- **Software Engineering:** Python, C++, Object-Oriented Programming (OOP), Git version control, Code reviews.
- **Artificial Intelligence:** Physics-Informed Neural Networks (PINNs), Self-Supervised Learning, Deep Learning (PyTorch, scikit-learn), U-Net architectures, and Attention Mechanisms.
- **Robotics & Systems:** ROS/ROS2, Linux (Ubuntu), Digital Twin (Siemens Tecnomatix), system testing, sensor calibration.
- **Leadership & Management:** 15+ years managing technical teams, R&D project lifecycles, and industrial innovation projects.

PROFESSIONAL EXPERIENCE

10/2024 – Actual

Postdoctoral Researcher

Queen's University | Kingston, Canada

- **Lead Developer for State Estimation:** Designing robust positioning algorithms using Invariant Kalman Filters (IEKF) to autonomous vehicles navigation in challenging environments, drone swarm cohesion and trajectory fidelity in communication-denied environments.
- **Deep Learning & Physics-Informed Networks:** Engineering high-performance state estimation frameworks by integrating physical constraints (vehicle dynamics, rotational consistency) and multi-sensor data into deep learning architectures to solve navigation tasks.
- **Full-Lifecycle Deployment:** Managing the integration of multi-sensor suites (Radar, LiDAR, GNSS, IMU) onto a ROS2-based self-driving platform for real-world validation.
- **Cross-Functional Leadership:** Coordinating a cross-functional team of graduate students and industry partners, driving innovation in multi-sensor autonomous navigation.
- **Hardware-Software Validation:** Manage the full lifecycle deployment of a self-driving research platform (ROS2/Linux), overseeing hardware-software integration of Radar, LiDAR, GNSS, IMU and other sensors for real-world testing. Platform demonstration: [Navinst Dataset](#)

08/2019 – 08/2020 Technical Director

SmartPLM | Sao Caetano do Sul, Brazil

- Directed the Digital Manufacturing division, leading the simulation and validation of complex industrial automation systems using Siemens Tecnomatix.
- Architected technical solutions for process simulation (Digital Twin), analyzing system kinematics and logic to optimize throughput and mitigate integration risks.
- Provided strategic technical guidance on system integration, ensuring alignment between simulation models and physical implementation.
- Collaborated with sales teams to develop new business opportunities and technical proposals.
- Supported marketing initiatives by advising on the technical aspects of campaigns and product launches.

06/2018 – 08/2021 Founder and Technical Director

Visium Labs Solutions | Sao Paulo, Brazil

- Designed and deployed custom Computer Vision and Machine Learning systems for industrial quality control and automation.
- Secured R&D funding (FAPESP) to develop proprietary vision algorithms, managing the project from concept to prototyping and validation.
- Designed and implemented machine vision systems tailored to industrial use cases, improving automation and quality control.
- Consultant on digital manufacturing and discrete event simulation projects, advising industrial clients on solution design and implementation.

01/2016 – 12/2017 Co-founder and Technical Director

Linesys cPDM Services | Santo Andre, Brazil

- Architected and developed immersive Virtual Reality (VR) and Augmented Reality (AR) applications for industrial process visualization, utilizing real-time rendering engines.
- Led technical pilot projects to validate emerging technologies in industrial environments, defining hardware requirements and software integration strategies for prospective use-cases.
- Collaborated with development teams to align software capabilities with technical customer requirements, bridging the gap between user needs and engineering constraints.

12/2010 – 05/2014 Presales Solutions Consultant

Siemens Industry Software | Sao Caetano do Sul, Brazil

- Served as a technical subject matter expert for Siemens Tecnomatix solutions, specializing in Discrete-Event Simulation, Robotics kinematics, and Automation validation.
- Developed complex technical prototypes and simulations to validate manufacturing processes, ensuring feasibility of automation logic and robotic reach/cycle times before physical deployment.
- Contributed to global technical forums by providing feedback on software features and developing custom prototype demonstrations to address specific engineering challenges.
- Delivered advanced technical training and troubleshooting for internal engineering teams and external stakeholders, ensuring successful adoption of simulation tools.
- Collaborated with sales teams to align technical strategy with client objectives and assist in closing business opportunities.
- Supported marketing teams by contributing technical insights for product campaigns and go-to-market strategies.

05/2008 – 11/2010 PLM Application Technician

TECMES | Sao Paulo, Brazil

- Executed technical implementations of Dassault Systèmes DELMIA suite, focusing on Robotics simulation, Ergonomics, and Process Planning validation.
- Resolved complex technical inquiries regarding software implementation and integration, creating custom training materials to enhance user proficiency in simulation workflows.
- Provided technical support to clients, resolving inquiries and guiding users in effective solution implementation.
- Collaborated with sales and technical teams to align demonstrations with customer needs and solution capabilities.

ACADEMIC BACKGROUND

09/2020 – 09/2024 Ph.D. in Electrical and Computer Engineering | Queen's University | Kingston, Canada

10/2019 – 07/2021 MBA in Business Management | University of Sao Paulo/ESALQ | Piracicaba, Brazil

01/2017 – 11/2018 M.Sc. in Eng. and Innovation Management | Federal University of ABC | Santo Andre, Brazil

05/2011 – 12/2016 Bachelor in Science and Technology | Federal University of ABC | Santo Andre, Brazil
Exchange scholarship in the **B.Sc. in Production Eng. and Management** program | University of Strathclyde | Glasgow, UK

08/2006 – 07/2009 Technologist in Production Processes | Faculty of Technology of Sao Paulo | Sao Paulo, Brazil

08/2006 – 07/2008 Mechatronics Technician | Faculty of Mechatronics SENAI | Sao Caetano do Sul, Brazil

SELECTED PUBLICATIONS

- QuasittudeNet: Self-Supervised Network for Vehicle Attitude Estimation. *IEEE Robotics and Automation Letters*. 10.1109/LRA.2026.3682616
- Robust Autonomous Vehicle Platooning in Communication-Denied Environments via Invariant Extended Kalman Filtering. *Accepted. 2026 IEEE 29th International Conference on Intelligent Transportation Systems (ITSC)*.
- The NavINST Dataset for Multi-Sensor Autonomous Navigation. 2025. *IEEE Access*. 10.1109/ACCESS.2025.3565878
- Comparative Study of Traditional and Deep Learning Feature Detectors and Matchers for Land Vehicle Monocular Visual Odometry. 2025. *2025 IEEE International systems Conference (SysCon)*. DOI: SysCon64521.2025.11014855
- Toward land vehicle ego-velocity estimation using deep learning and automotive radars. 2024. *IEEE Transactions on Radar Systems*. DOI: 10.1109/TRS.2024.3392439
- A novel method for land vehicle positioning: invariant Kalman filters and deep-learning-based radar speed estimation. 2023. *IEEE Transactions on Intelligent Vehicles*. DOI: 10.1109/TIV.2023.3287790
- Smart Mobility: Leveraging perception sensors for map-based navigation in autonomous vehicles. 2024. *2024 IEEE International Conference on Smart Mobility (SM)*. DOI: 10.1109/SM63044.2024.10733404
- Robust land vehicle positioning with invariant Kalman filters and map-based corrections. 2023. *2023 IEEE 26th International Conference on Intelligent Transportation Systems (ITSC)*. DOI: 10.1109/ITSC57777.2023.10422638
- Evaluation of invariant extended Kalman filters applied to multi-sensor land vehicle navigation in GNSS challenging environments. 2022. *Proceedings of the 35th International Technical Meeting of the Satellite Division of The Institute of Navigation (ION GNSS+ 2022)*. DOI: 10.33012/2022.18472